The course materials cover a structured set of machine learning models, feature selection techniques, and core theoretical concepts:

1. Types of Models

- **Regression Models**: Includes **Linear Regression**, **Logistic Regression**, **Ridge Regression (L2)**, **Lasso Regression (L1)**, and **Elastic Net Regression**.
- **Support Vector Machines (SVM)**: Covers **Hard Margin SVM**, **Soft Margin SVM**, and **Kernel SVM** using Linear, Polynomial, and RBF kernels.
- **Neural Networks**: Focuses on **Artificial Neural Networks (ANN)** and **Multilayer Perceptron (MLP)** for classification and regression tasks.
- **Decision Trees**: Details **Classification Trees** (using Gini Impurity) and **Regression Trees** (using Sum of Squared Residuals).
- **Ensemble Models**: Combines weak learners into strong learners using:
    - **Bagging (Bootstrap Aggregating)**: e.g., **Random Forest**.
    - **Boosting**: e.g., **Gradient Boost**.
    - **Stacking**: Training a meta-model on predictions from base models.
- **Reinforcement Learning**: Features **Q-Learning** (agent, environment, state, action, reward).
- **Clustering Models**: Covers **K-Means Clustering** and **Hierarchical Clustering**.

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2. Feature Selection Methods

Feature selection is divided into three primary categories:

- **Embedded Methods (Regularization)**: Features are selected as part of model optimization, including **Ridge Regression (L2)**, **Lasso Regression (L1)**, and **Elastic Net Regression**.
- **Filter Methods**: Evaluates individual feature characteristics, including **Correlation (Pearson's R)**, **Adapted Lasso**(filtering features by non-zero initial coefficients), and **ANOVA (F-test)**.
- **Wrapper Methods**: Evaluates groups of features using a machine learning model, specifically **Recursive Feature Elimination (RFE)** via **Forward** and **Backward** selection.

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3. Other Key Concepts

- **Dimensionality Reduction Techniques**: Includes **Principal Component Analysis (PCA)**, **t-SNE** (t-distributed stochastic neighbor embedding), **Random Projections**, and **Feature Agglomeration**.
- **Validation & Data Partitioning**: **Training**, **Validation**, and **Test sets**, along with **K-Fold Cross Validation**.

- **Performance Evaluation Metrics**: **Accuracy**, **Confusion Matrix**, **Precision**, **Recall**, **F1-Score**, and **ROC & AUC curves**.
- **Loss / Cost Functions**: **Huber Loss**, **Log-Cosh Loss**, **Quantile Loss**, **Binary Cross Entropy**, **Categorical Cross Entropy**, and **Mean Squared Error (MSE)**.
- **Learning Principles & Mechanics**: **Bias-Variance Tradeoff**, **Neural Node structure** (weights, bias, activation functions, backpropagation), and **Proximity / Distance Measurements**.

💡 Would you like me to generate a summary report or a structured comparison table comparing the Embedded, Filter, and Wrapper feature selection methods?